
Taipei River Pulse : Interactive system for water level warning and attraction exploration in five major water systems

Revision History

Revision 4

- Added AI-supported proof in the last chapter
- 2026-06-16
- Group O : 412402141 陳怡珊 、 412402452 葉庭蓁

Revision 3

- Added the fourth subsystem, text corrections, and UI images
- 2026-06-11
- Group O : 412402141 陳怡珊 、 412402452 葉庭蓁

Revision 2

- Text corrections and UI images
- 2026-06-07
- Group O : 412402141 陳怡珊 、 412402452 葉庭蓁

Revision 1

- 2026-06-03
- Group O : 412402141 陳怡珊 、 412402452 葉庭蓁

System Specification

Requirement Definition

Taipei River Pulse is a web-based WebGIS system designed to visualize and explore the five major river systems in the Greater Taipei area, including Tamsui River, Dahan River, Xindian River, Jingmei River, and Keelung River. The system integrates open-source GIS technologies, interactive map visualization, real-time water-level data, cultural tourism content, location-based services, and flood-warning information.

The system is developed as a lightweight front-end WebGIS platform. It uses GeoJSON as the primary spatial data format and Leaflet.js as the map rendering engine. River routes, cultural points of

interest, badge collection areas, water-risk zones, and monitoring station markers are displayed as interactive map layers. In addition, Firebase Authentication and Firestore are used to support Google login, cloud synchronization, personal records, and leaderboard services.

The main objectives of the system are : (1) to provide a river-oriented spatial exploration platform instead of an administrative-area-based map, (2) to integrate real-time water-level information from government open data, (3) to support cultural tourism through river-based storytelling and Google Street View heritage guidance, (4) to provide attraction search, category filtering, and interactive cultural quizzes, (5) to support flood-risk awareness through dynamic water-level status and warning messages, (6) to provide location-based interaction including GPS positioning, nearby attraction detection, badge collection, and personalized exploration records, (7) to provide user authentication and cloud synchronization through Firebase Authentication and Firestore, enabling cross-device access and leaderboard services.

Development Environment

- Target web browsers:
 - Google Chrome
 - Microsoft Edge
 - Apple Safari

- Development tools
 - Frontend Development : HTML5, CSS3, JavaScript
 - Data Sources:
 - GeoJSON: Standard format for encoding geographic data structures.
 - Government Open Data API: External data integration for real-time information.
 - Web-based GIS map view
 - Leaflet is one of the most popular open-source JavaScript libraries for interactive maps.
 - OpenStreetMap tile layer: Base map data source.
 - Google Maps Street View Embed: Integration for 360-degree street views.
 - Cloud Services:
 - Firebase Authentication: Google account login and user authentication.
 - Firebase Firestore: Cloud storage for user profiles, badge collections, quiz scores, river exploration records, and leaderboard data.
 - Environment and system test
 - Visual Studio Code: Code editor (IDE)
 - Live Server / ngrok for mobile testing: Local hosting and secure tunnels for remote mobile testing.

- **Deploy a website with GitHub Pages :** <https://shan11111111.github.io/taipei-river-system>

- **Data Standard**

- Follow ISO/TC211 and OGC standards (e.g., GeoJSON) to process related metadata and spatial datasets.
- Conform to the coordinate systems in Taiwan (utilizing WGS 84 / EPSG:4326 for geospatial referencing)

System Scope

Taipei River Pulse includes 4 subsystems as below:

1. River Spatial Visualization Subsystem

This subsystem manages the visualization of the five river systems. Each river is displayed with a different color and animated flow effect. Users can hover over or click a river to highlight it and view its basic description. Main functions include:

- A. Five major river route visualization
- B. River hover and click interaction
- C. Animated river flow effect
- D. River layer control
- E. Historical river route demonstration layer
- F. Flood-risk buffer visualization

2. Cultural Tourism and Location-Based Guide Subsystem

This subsystem provides cultural and tourism information along the river corridors. Points of interest are displayed on the map and can be opened to show historical stories, cultural explanations, and Google Street View-based heritage guidance. Main functions include:

- A. Cultural POI markers
- B. Heritage story cards
- C. Google Street View guide panel
- D. River-perspective storytelling
- E. GPS positioning
- F. Nearby POI detection
- G. Badge collection mechanism
- H. Badge collection interface

3. Real-Time Water-Level Monitoring Subsystem

This subsystem connects to the Water Resources Agency Open Data API and retrieves real-time water-level data. The system maps representative monitoring stations to the five river systems and displays the latest available water level and warning level. Main functions include:

- A. Real-time water-level API integration
- B. Representative station mapping
- C. Safe / warning status display
- D. Automatic data refresh every minute
- E. Manual refresh button
- F. Simulated warning mode for demonstration
- G. Flood warning pop-up notification

Note: The system queries the API every minute, but the observation time depends on the update frequency of the government data source.

4. User Profile, Cloud Sync, and Leaderboard Subsystem

This subsystem manages user authentication, cloud synchronization, personal exploration records, and the leaderboard system. Users can sign in with their Google accounts, synchronize their progress across different devices, collect badges, and compare their achievements with other users while maintaining privacy settings. Main functions include:

- A. Google account authentication
- B. Firebase cloud synchronization
- C. Personal profile management
- D. Badge collection history
- E. Quiz score synchronization
- F. River exploration record synchronization
- G. Leaderboard ranking
- H. Display name privacy setting
- I. Export and clear personal records

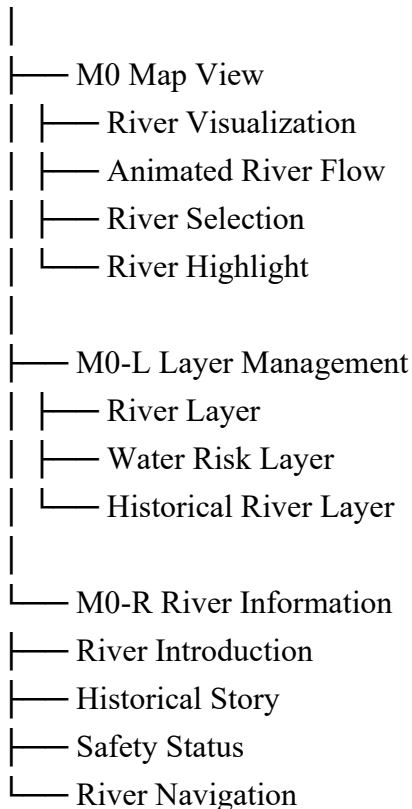
System Development Plan

The system development plan focuses on developing an interactive WebGIS platform for visualizing the five major river systems in Greater Taipei. The system integrates spatial visualization, real-time water-level monitoring, cultural tourism guidance, location-based services, and cloud-based user management through Firebase Authentication and Firestore.

River Spatial Visualization Subsystem

System Structure

River Spatial Visualization Subsystem



Conceptual UI Elements

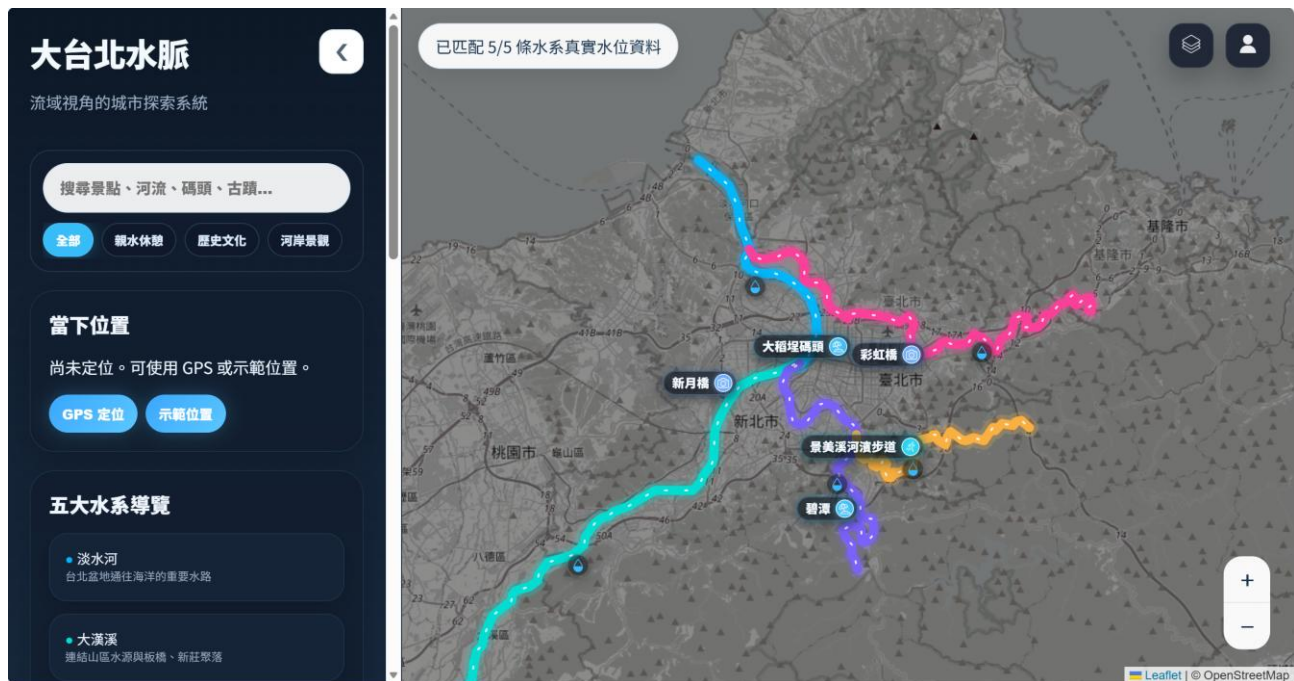
- (1) The welcome screen



(2) Watch the 30-second introduction animation or skip it.



(3) Our main page of our system



(4) The left sidebar interface





Main Functions

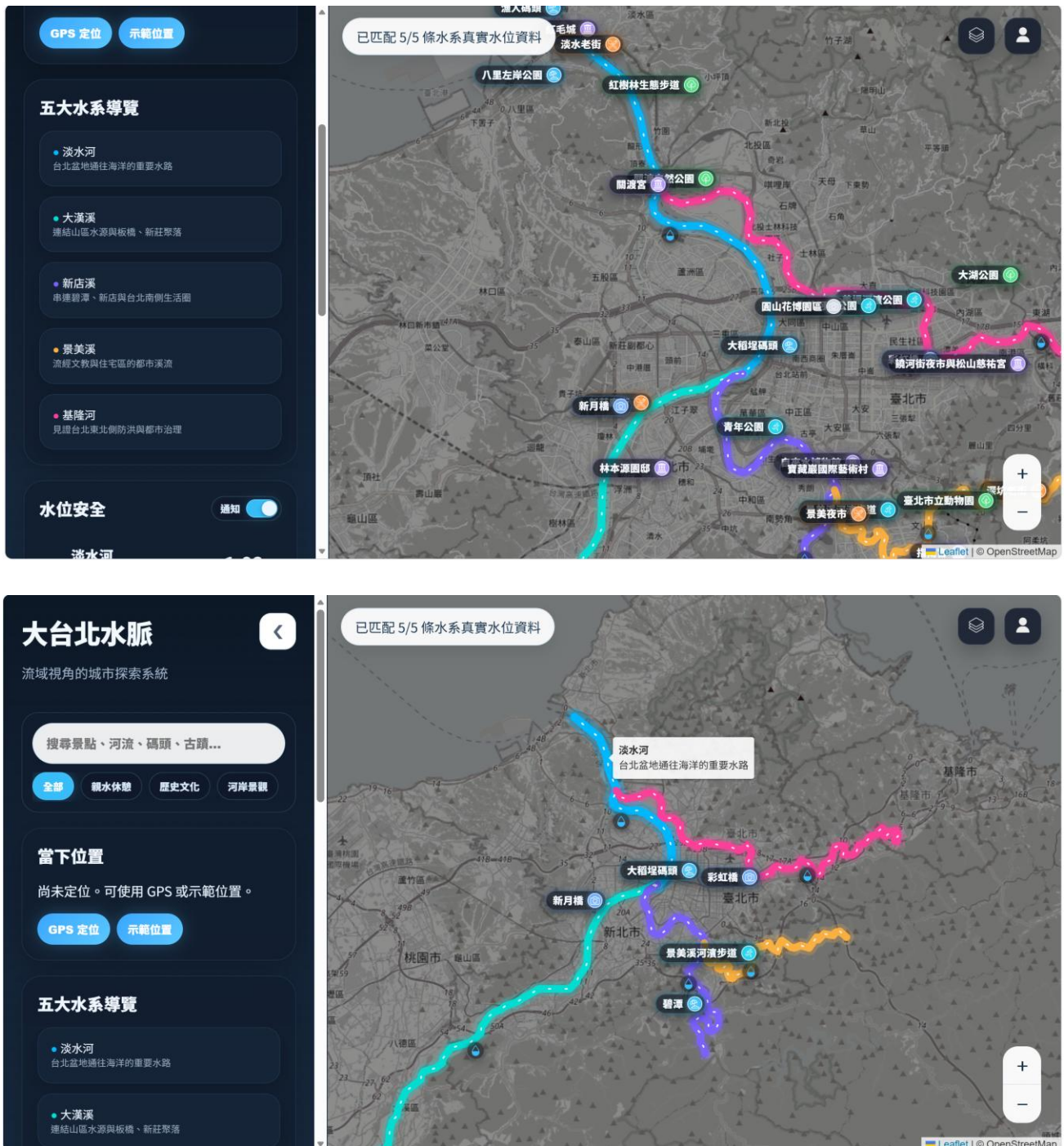
The main task of this subsystem is to visualize the five major river systems and provide interactive river-based map exploration

1. Map View (M0)

The main map view displays the Greater Taipei area and the five major river systems. Users can browse the map, select rivers, view POI markers, and access water-level information.

● UI Elements

M0-01	River selection panel
M0-02	Main WebGIS map view
M0-03	River hover highlight
M0-04	River click interaction
M0-05	Current mode display



2. Custom Overlays 圖層管理 (M0-L)

This view allows users to control the visibility of different map layers..

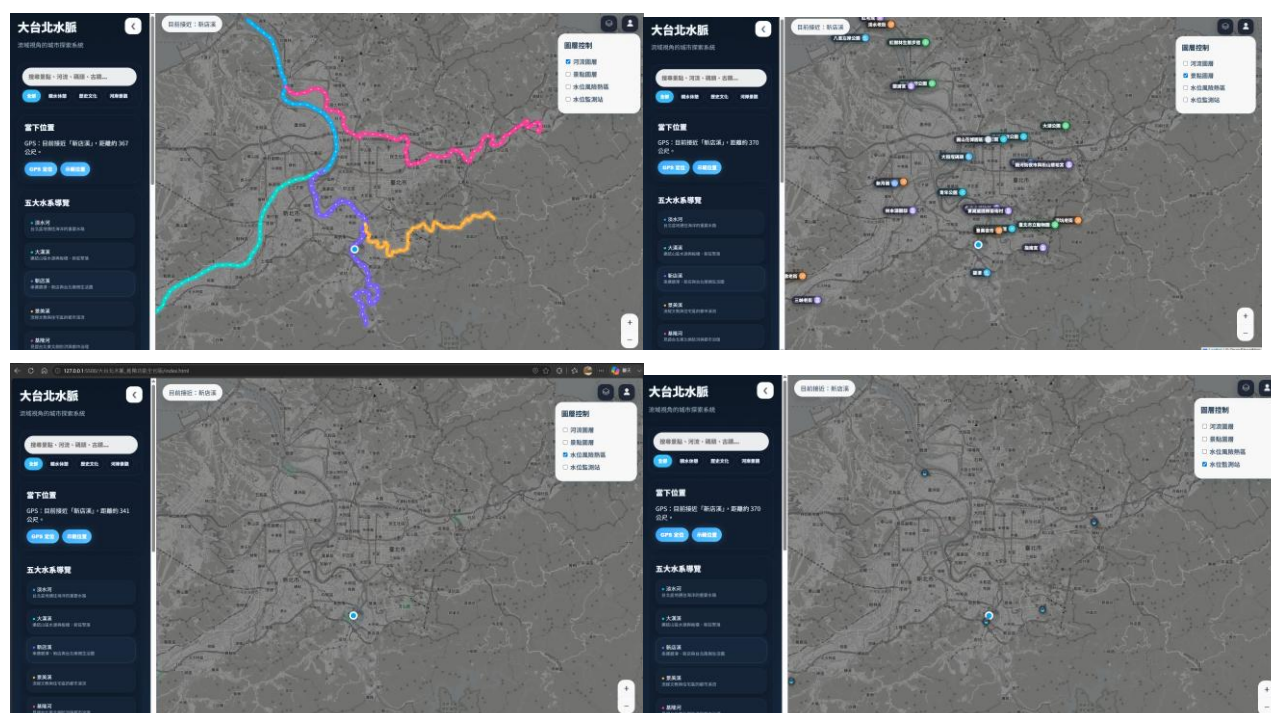
- Available independent layers:

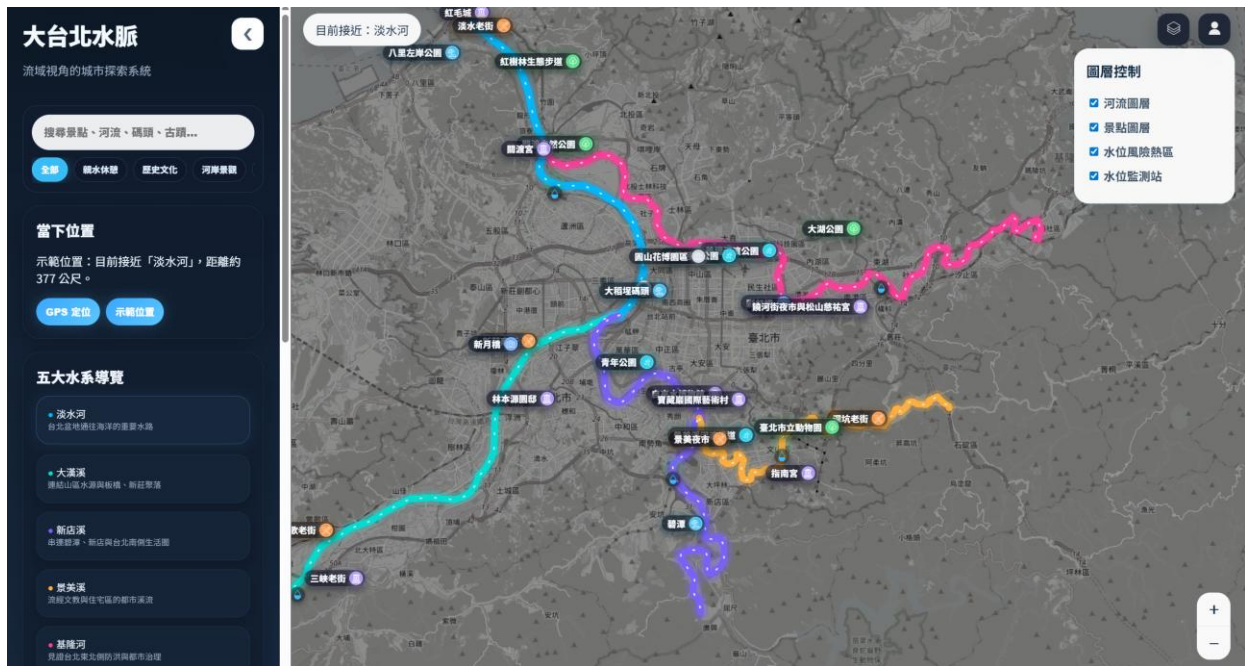
Layer Name	Description
Available mutually exclusive layers: (Exactly one of these layers can be active at any time.)	
River Layer	Five major river systems
POI Layer	Cultural and tourism attractions

Water Risk Layer	Water-level warning zones
Monitoring station layer	Signal monitoring station

UI Elements

M0-L01	River layer toggle
M0-L02	POI layer toggle
M0-L03	Water risk layer toggle
M0-L04	monitoring station toggle



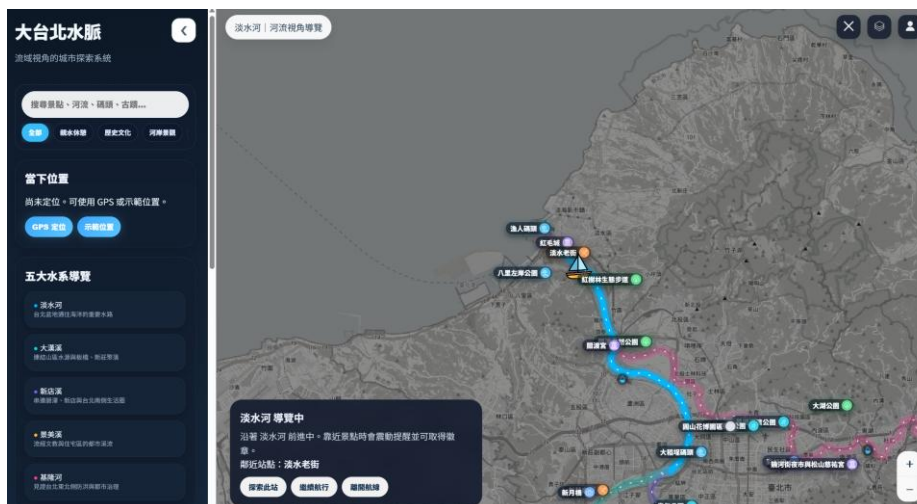
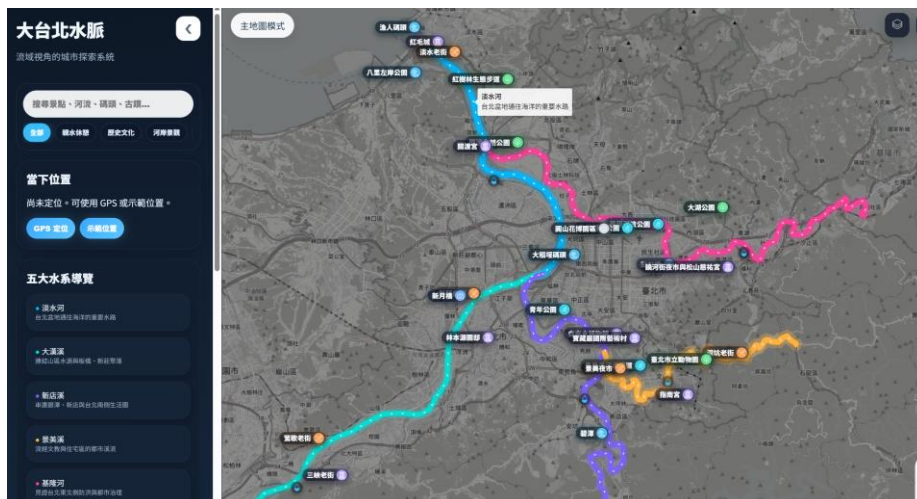


3. River Information (M0-R)

This view appears when users click a river. The system displays river information, historical background, and current water conditions.

● UI Elements

M0-R01	River name
M0-R02	River description
M0-R03	Historical story
M0-R04	Water safety status
M0-R05	Enter river navigation button
M0-R06	Voice introduction button



Cultural Tourism and Location-Based Guide Subsystem

System Structure

Cultural Tourism and Location-Based Guide Subsystem

|

- M0-P POI Guide
 - Attraction Information
 - Heritage Story
 - Cultural Quiz
 - Street View Guide
 - River Perspective Storytelling
- M0-S POI Search and Filter
 - Keyword Search
 - Category Filter
 - Search Result List
- M0-G GPS Navigation
 - Current Location
 - Nearby Attraction Detection
 - Distance Display
- M0-B Badge Collection
 - Badge Collection
 - Collection Status
 - Badge Board

Conceptual UI Elements



Main Functions

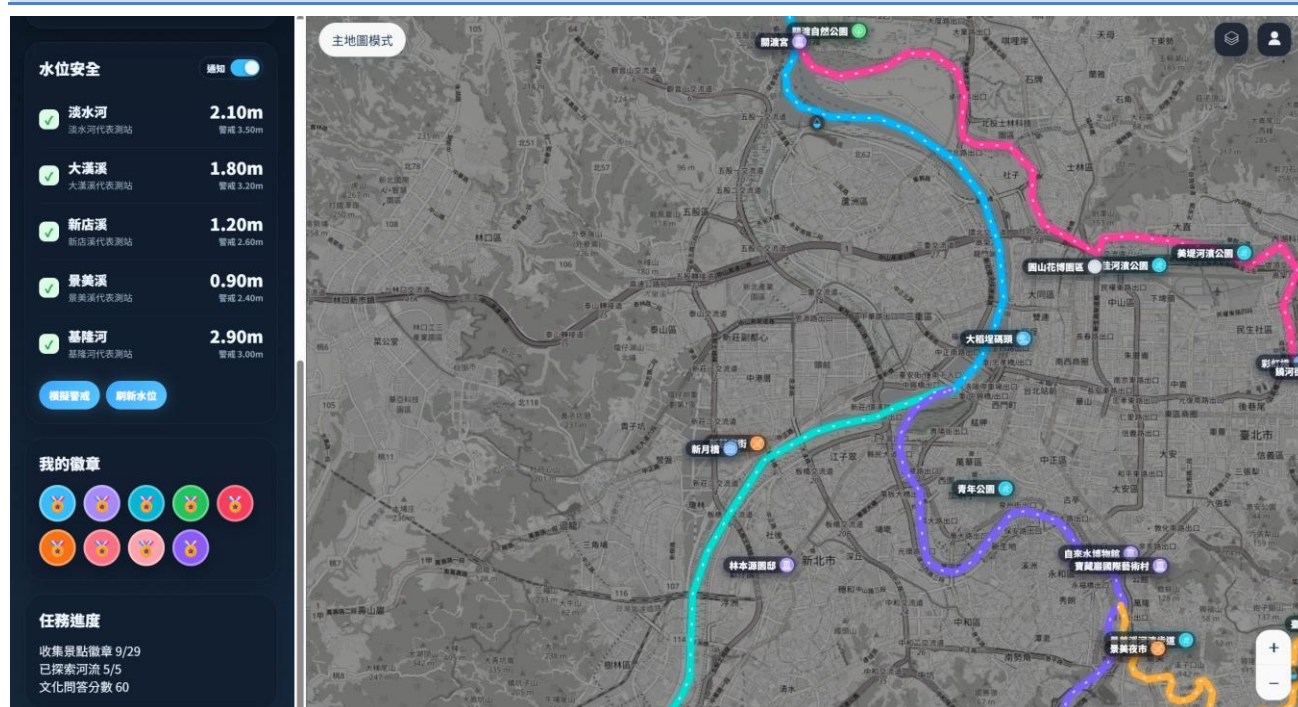
The main task of this subsystem is to provide cultural tourism guidance, POI exploration, location-based interaction, and badge collection.

1. POI Guide (M0-P)

This view appears when users click a cultural attraction marker. It displays attraction information, cultural stories, and related quiz content.

● UI Elements

M0-P01	Attraction name
M0-P02	Attraction story
M0-P03	Cultural quiz area
M0-P04	Badge collection status
M0-P05	POI marker



GOOGLE STREET VIEW / HERITAGE GUIDE
景美夜市 | 街景古蹟導覽

開啟 Google Maps

12 溪口街23巷
臺北, 臺北市
透過 Google 地圖查看

街景導覽 河流視角 古蹟文化

景美夜市位於景美溪生活圈附近，展現河岸周邊居民日常飲食與夜間商業活動。

古蹟/景點：景美夜市與生活圈
 河流視角故事：景美夜市是文山區重要夜市，能呈現景美溪周邊居民生活與地方商圈文化。
 水位安全：✅ 景美溪 目前水位 0.90m，低於警戒值。

景美溪河濱與文教生活圈

GOOGLE STREET VIEW / HERITAGE GUIDE
景美溪河濱步道 | 街景古蹟導覽

開啟 Google Maps

10 羅斯福路6段166巷8弄
臺北, 臺北市
透過 Google 地圖查看

景美溪河濱與文教生活圈

景美溪流經住宅、學校與河濱步道，是日常型水岸空間。它不像港口那樣壯觀，卻能呈現河流如何進入居民生活。

水位安全：✅ 景美溪 目前水位 0.90m，低於警戒值。

指南宮與木柵山區水系

指南宮位於木柵山區，是台北重要信仰景點。放入景美溪導覽可讓使用者理解溪流上游不只是自然環境，也連結山區聚落與宗教文化。

臺北市立動物園與環境教育

臺北市立動物園位於木柵地區，鄰近景美溪流域，可作為環境教育與親子旅遊節點，讓水系導覽不只談歷史，也能談生態與教育。

GOOGLE STREET VIEW / HERITAGE GUIDE
指南宮 | 街景古蹟導覽

開啟 Google Maps

指南宮
透過 Google 地圖查看

景美溪流經住宅、學校與河濱步道，是日常型水岸空間。它不像港口那樣壯觀，卻能呈現河流如何進入居民生活。

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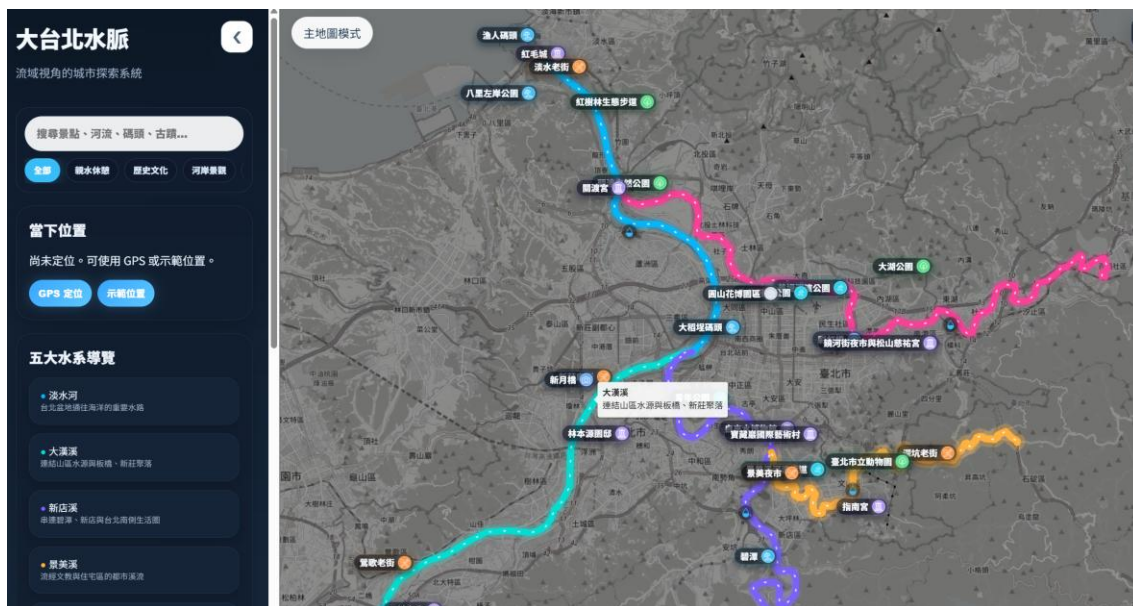
2. POI Search and Filter (M0-S)

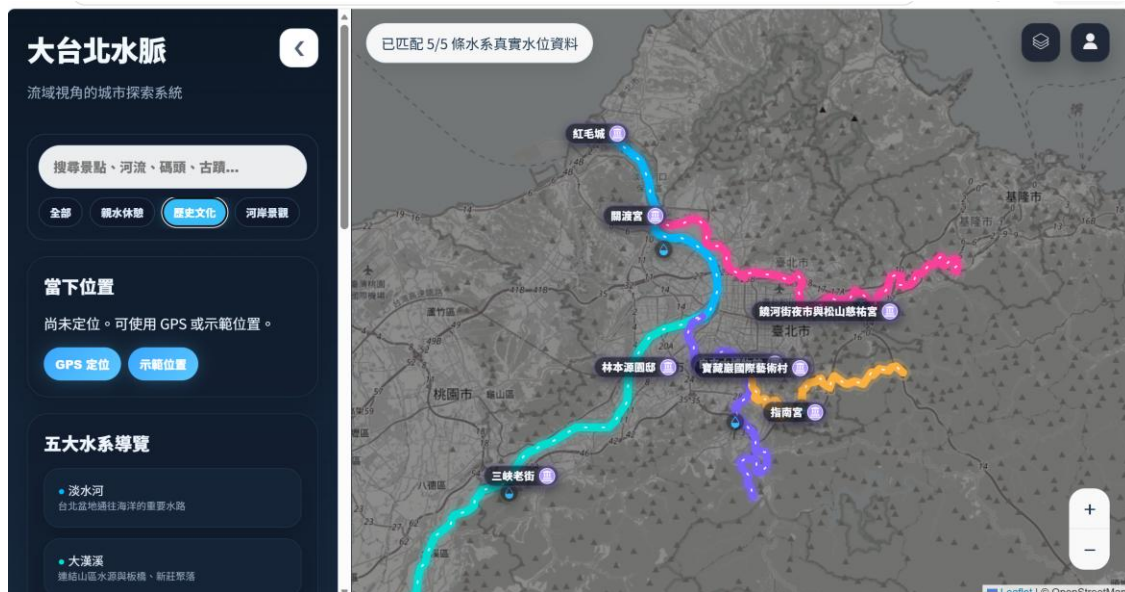
This function allows users to search and filter attractions by keyword or category.

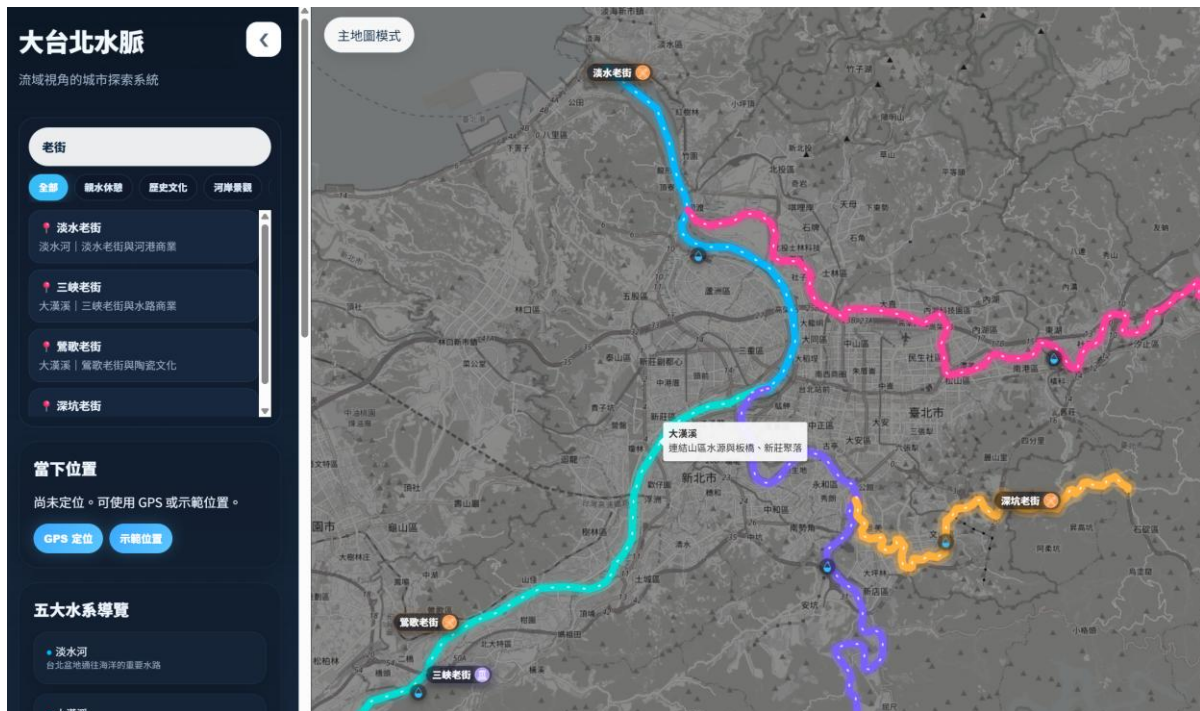
● UI Elements

M0-S01	POI search input
M0-S02	POI category filter
M0-S03	Search result list







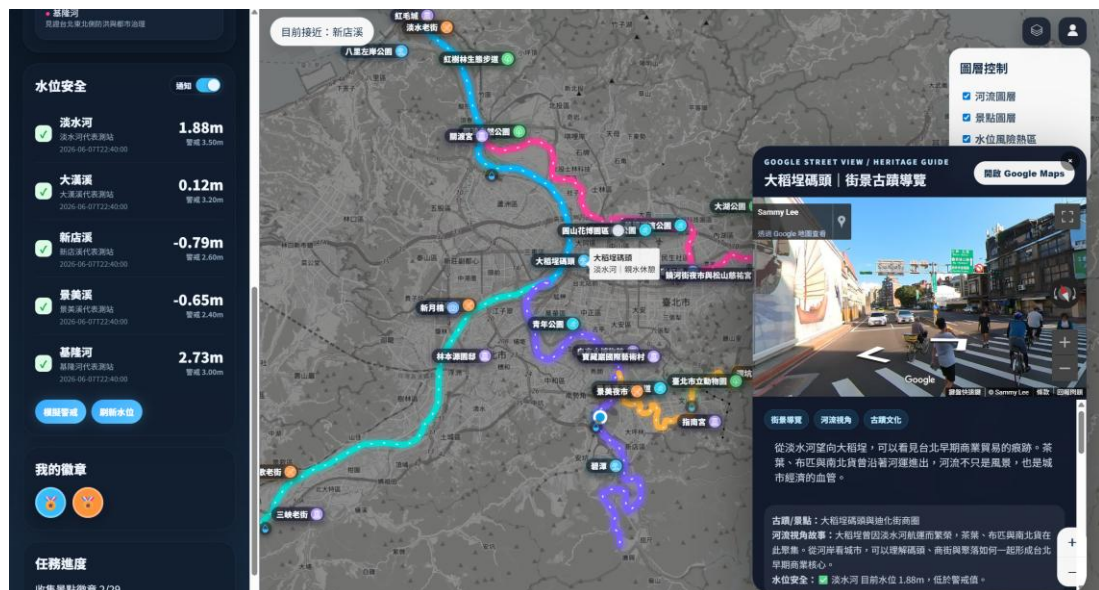


3. Street View Heritage Guide (M0-V)

This function provides Google Street View-based heritage guidance and river-perspective storytelling.

● UI Elements

M0-V01	Street View panel
M0-V02	Street View iframe
M0-V03	Heritage guide title
M0-V04	River-perspective story
M0-V05	Open Google Maps button

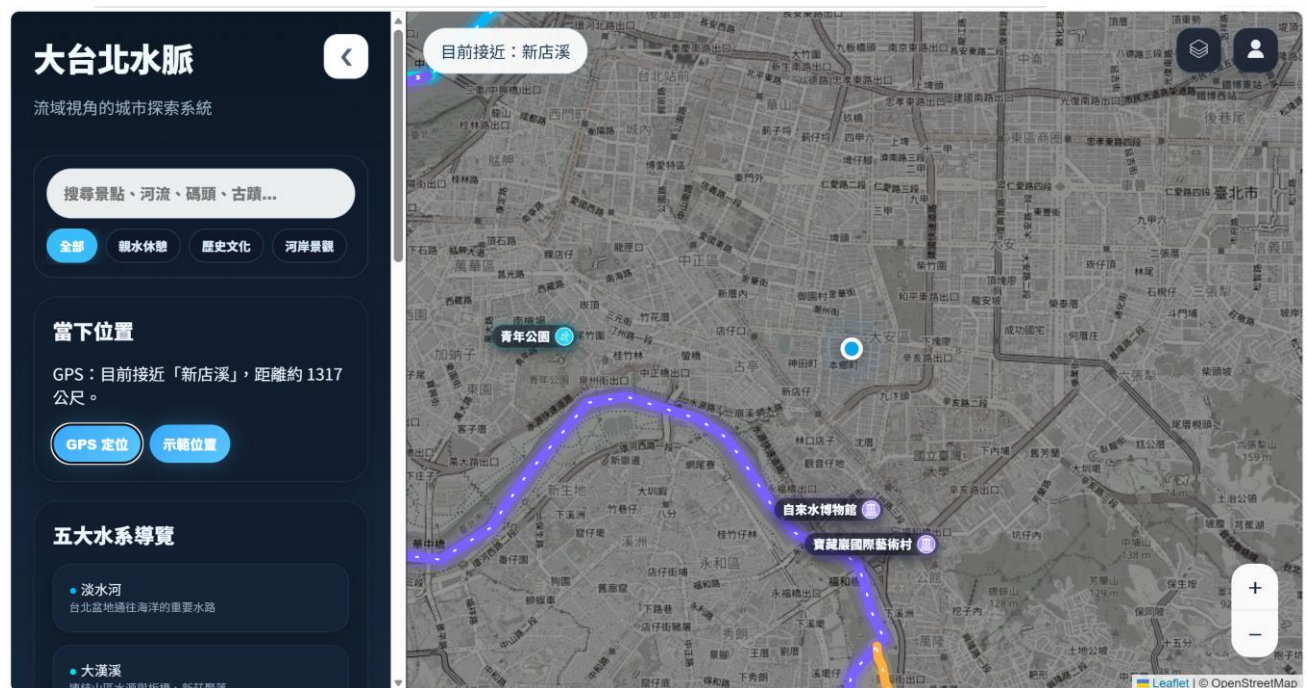






4. GPS Navigation (M0-G)

M0-G01	GPS location button
M0-G02	Current location text
M0-G03	Current location marker
M0-G04	Nearby attraction detection



5. Badge Collection (M0-B)

M0-B01	Badge board
M0-B02	Badge collection result
M0-B03	Badge status display
M0-B04	Cultural quiz completion result

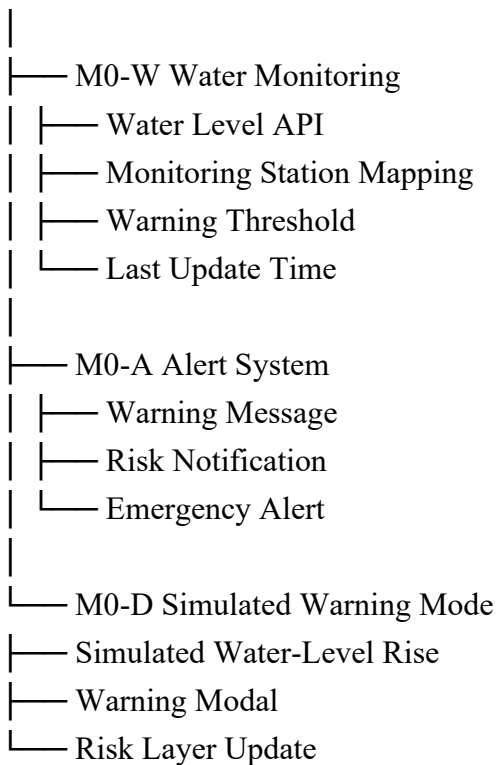




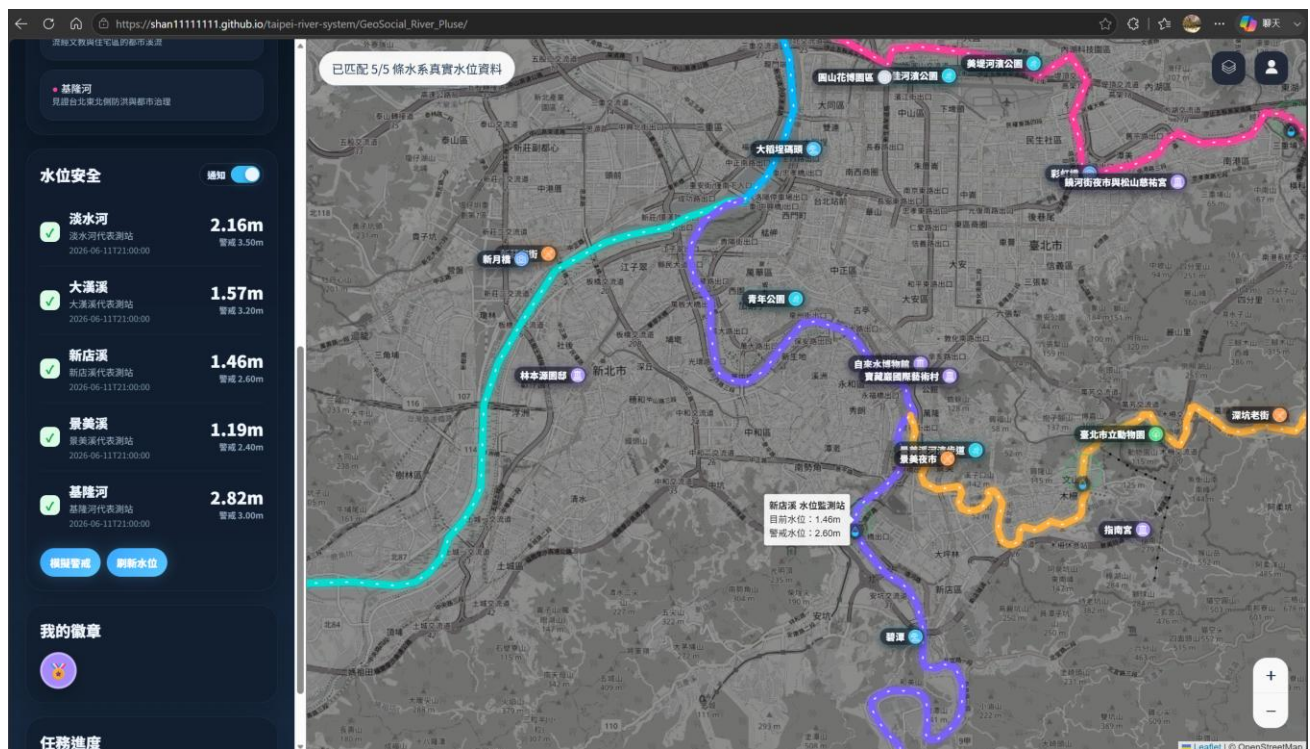
Real-Time Water-Level Monitoring Subsystem

System Structure

Water-Level Monitoring Subsystem



Conceptual UI Elements



Main Functions

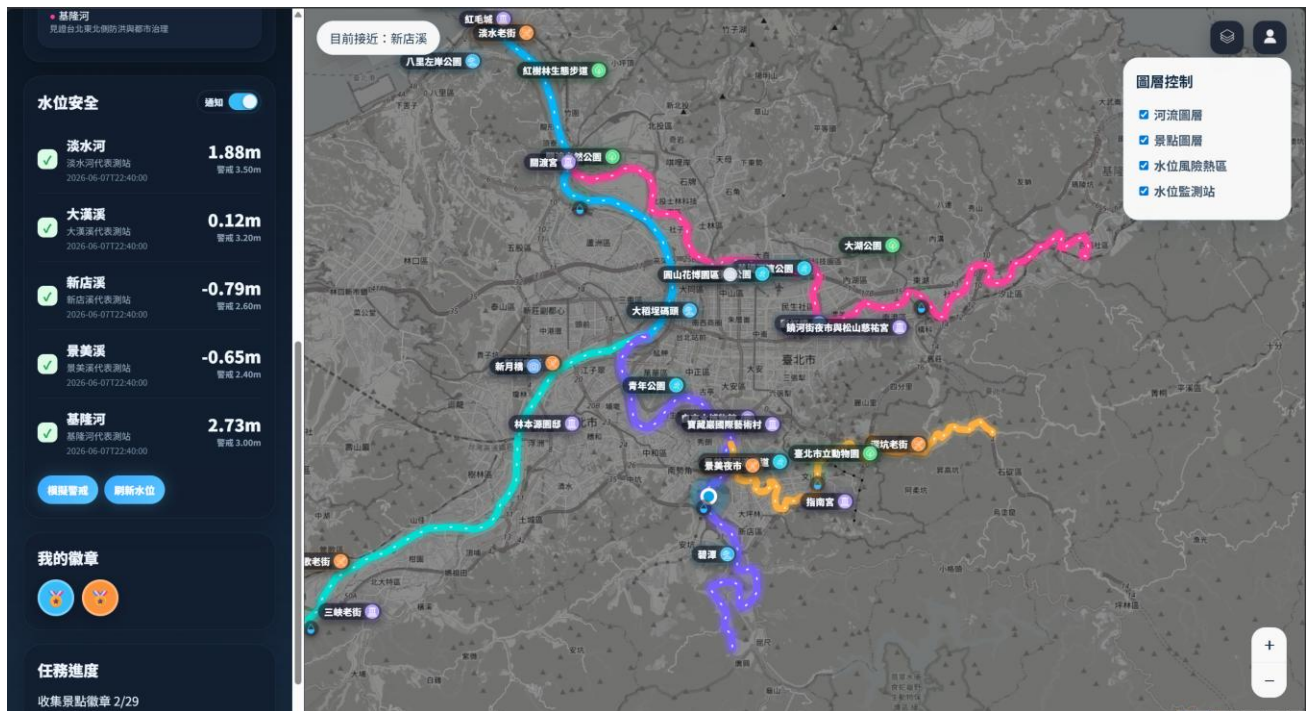
The main task of this subsystem is to retrieve real-time water-level information, display safety status, and provide flood-warning notifications.

1. Water Monitoring (M0-W)

This function retrieves water-level data from the Water Resources Agency Open Data API and displays the latest available monitoring information.

● UI Elements

M0-W01	River name
M0-W02	Monitoring station
M0-W03	Current water level
M0-W04	Warning level
M0-W05	Last update time
M0-W06	Refresh button

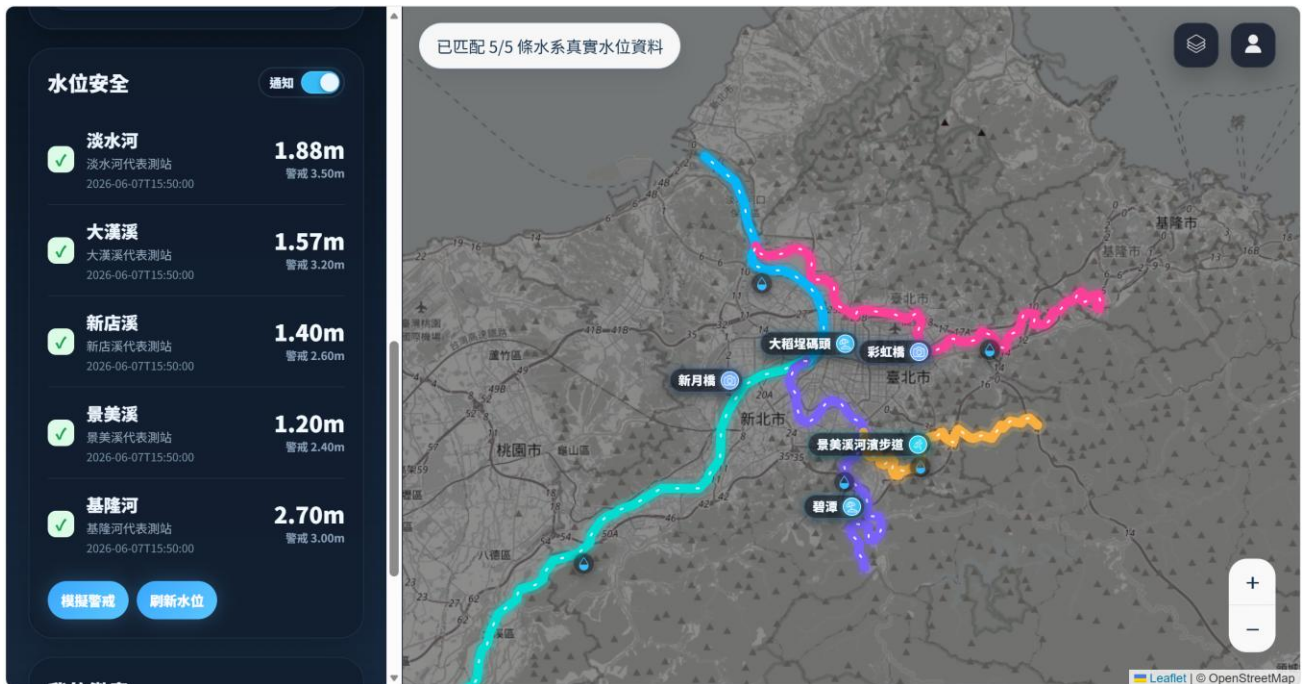


2. Flood Warning Alert (M0-A)

This function displays warning notifications when the water level reaches or exceeds the warning threshold.

● UI Elements

M0-A01	Water warning toast
M0-A02	Danger warning modal
M0-A03	Warning message text
M0-A04	Confirm warning button
M0-A05	Notification toggle switch

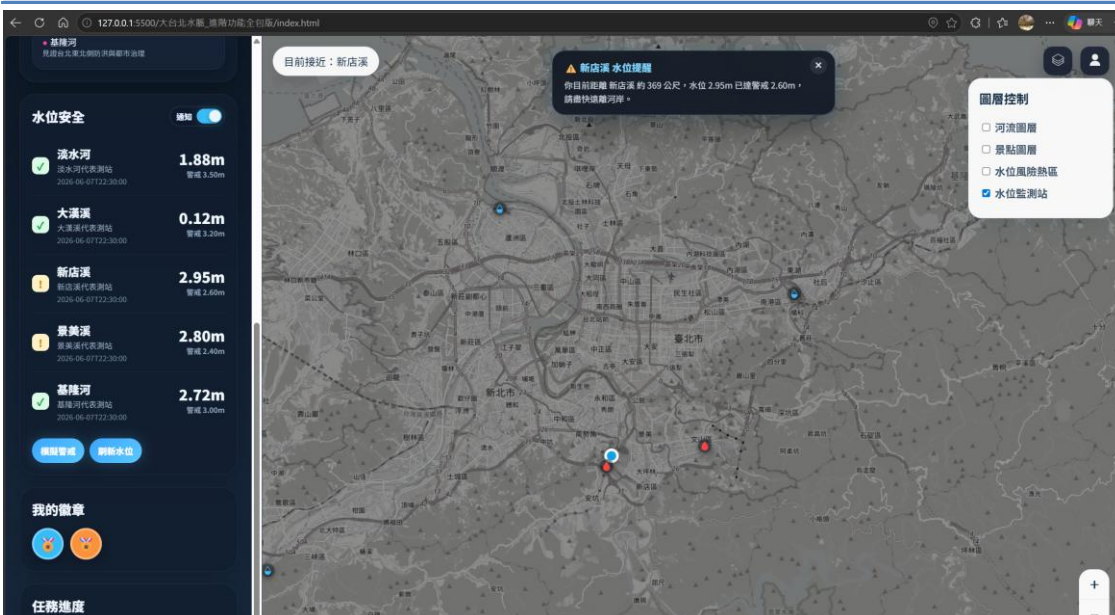


3. Simulated Warning Mode (M0-D)

This function provides a demonstration mode for flood-warning scenarios during classroom presentation.

● UI Elements

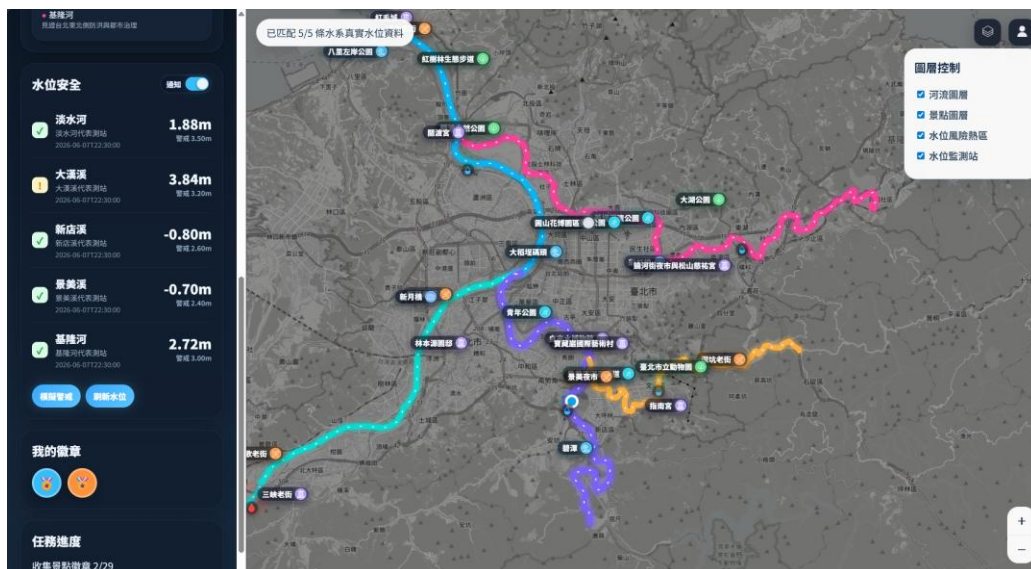
M0-D01	Simulate danger button
M0-D02	Simulated warning message
M0-D03	Risk layer update



▲ 水位警戒提醒

大漢溪 目前水位升高至 3.8m，已超過警戒水位 3.2m。請避免靠近河岸、低窪步道與親水區域。

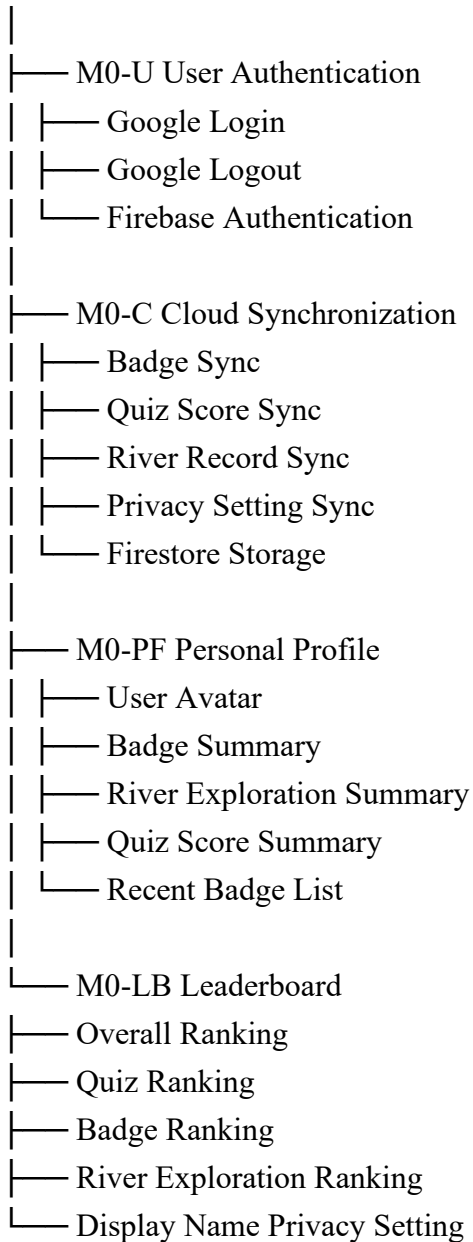
我知道了



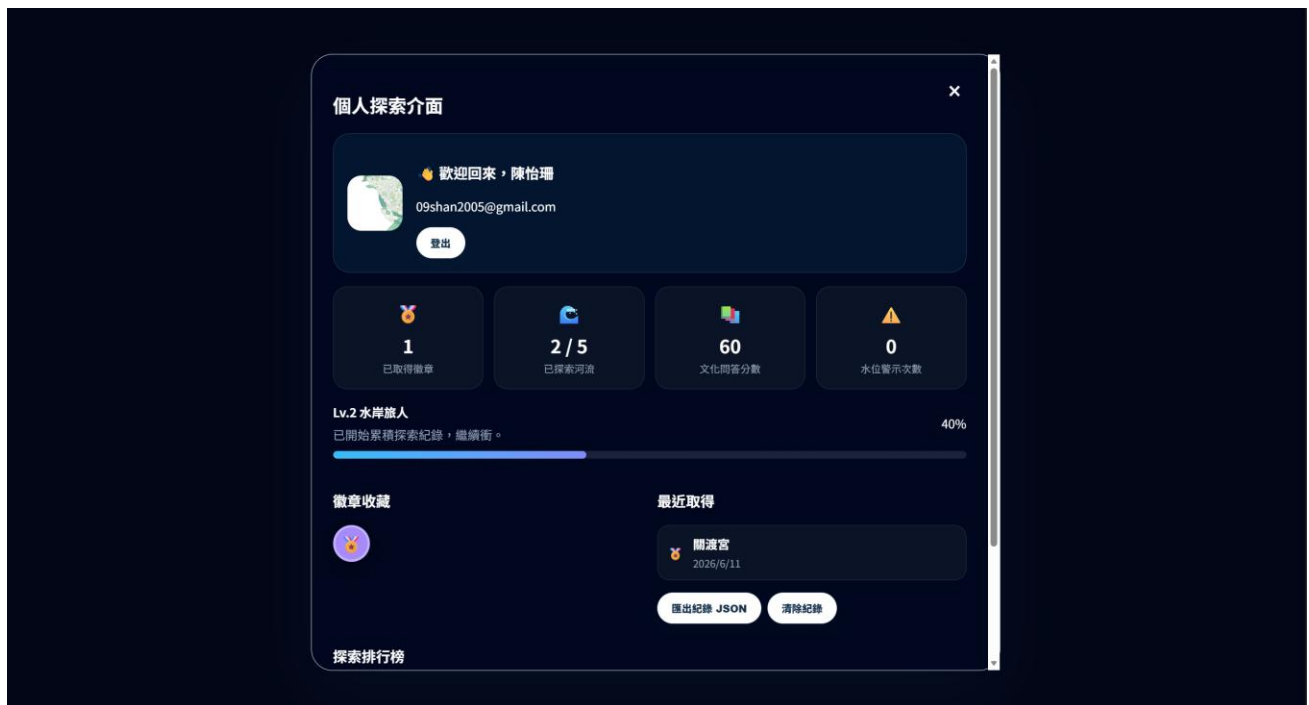
水位安全		通知 ☒
☒	淡水河 淡水河代表測站 2026-06-07T15:50:00	1.88m 警戒 3.50m
☒	大漢溪 大漢溪代表測站 2026-06-07T15:50:00	1.57m 警戒 3.20m
☒	新店溪 新店溪代表測站 2026-06-07T15:50:00	1.40m 警戒 2.60m
☒	景美溪 景美溪代表測站 2026-06-07T15:50:00	1.20m 警戒 2.40m
☒	基隆河 基隆河代表測站 2026-06-07T15:50:00	2.70m 警戒 3.00m
模擬警戒		刷新水位

System Structure

User Profile, Cloud Sync, and Leaderboard Subsystem



Conceptual UI Elements



Main Functions

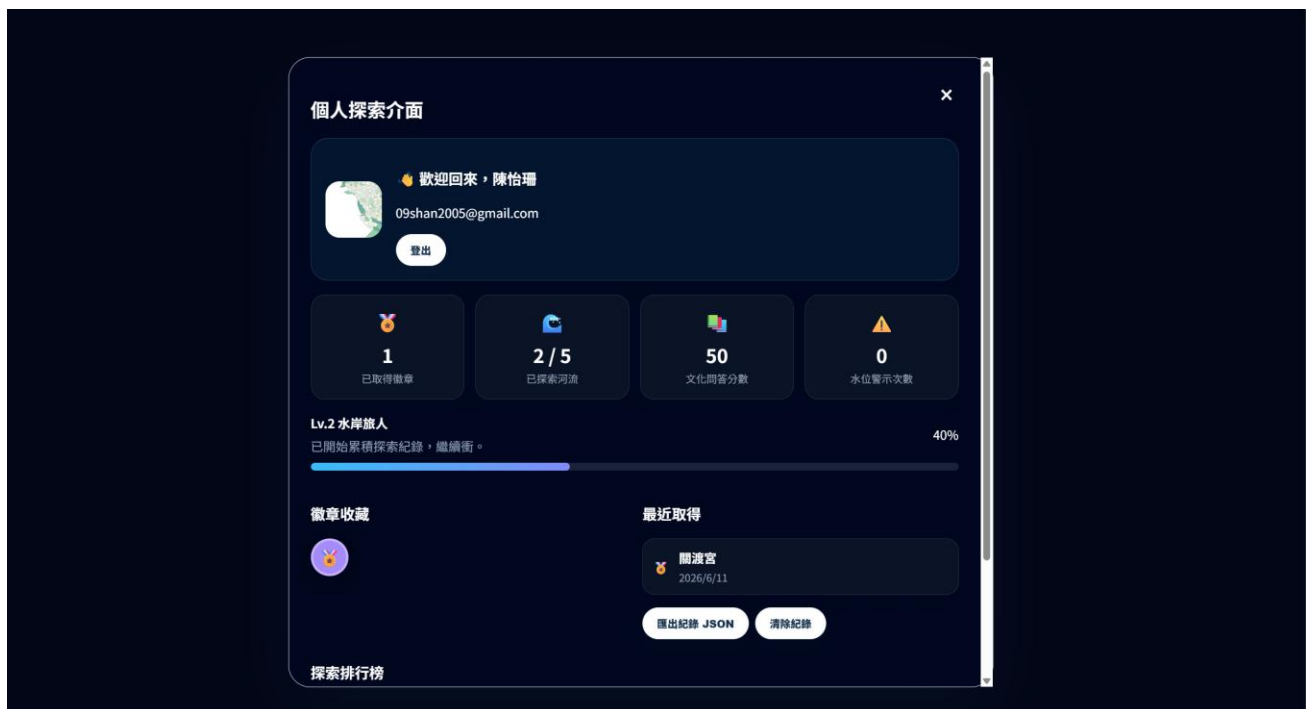
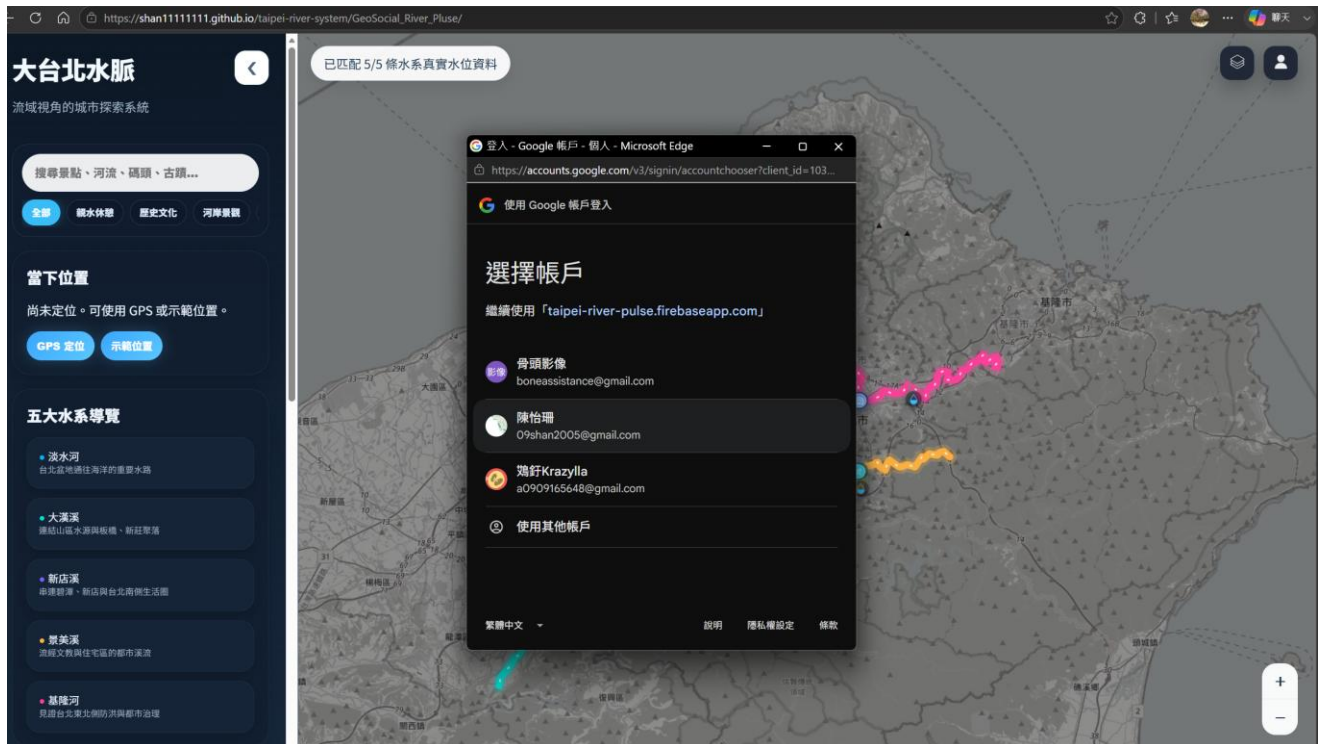
The main task of this subsystem is to provide user account management, cloud synchronization, and social interaction through the leaderboard.

1. User Authentication (M0-U)

Users can sign in using Google accounts through Firebase Authentication. Authentication enables cloud synchronization and leaderboard participation.

● UI Elements

M0-U01	Google login button
M0-U02	Google logout button
M0-U03	User avatar
M0-U04	User information



2. Cloud Synchronization (M0-C)

This function synchronizes badge collections, quiz scores, and river exploration records to Firebase Firestore.

● UI Elements

M0-C01 Badge synchronization

M0-C02	Quiz score synchronization
M0-C03	River record synchronization
M0-C04	Privacy setting synchronization
M0-C05	Cloud storage status

3. Personal Profile (M0-PF)

- UI Elements

M0-PF01	Profile user box
M0-PF02	Badge count
M0-PF03	Explored river count
M0-PF04	Quiz score
M0-PF05	Warning count
M0-PF06	Recent badge list
M0-PF07	Export record button
M0-PF08	Clear record button



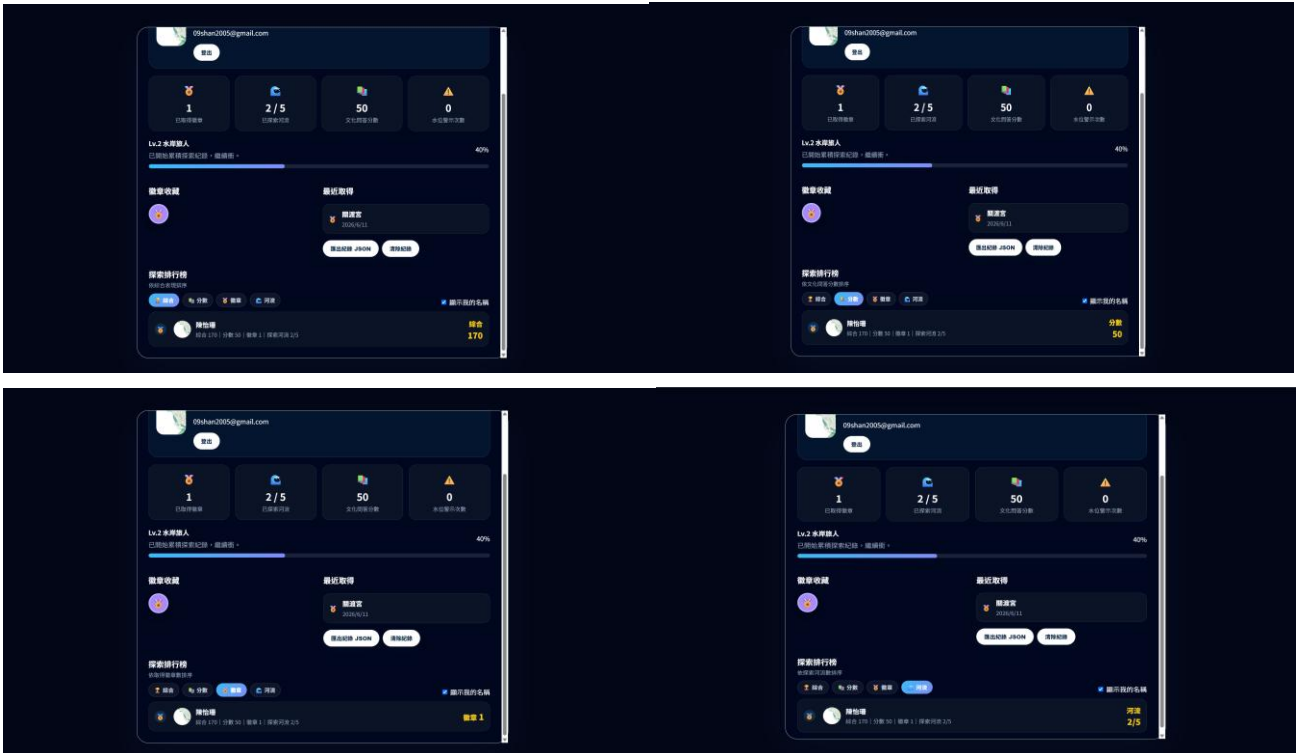
4. Leaderboard (M0-LB)

The leaderboard displays user achievements and supports multiple ranking modes.

- UI Elements

M0-LB01	Leaderboard list
M0-LB02	Overall ranking

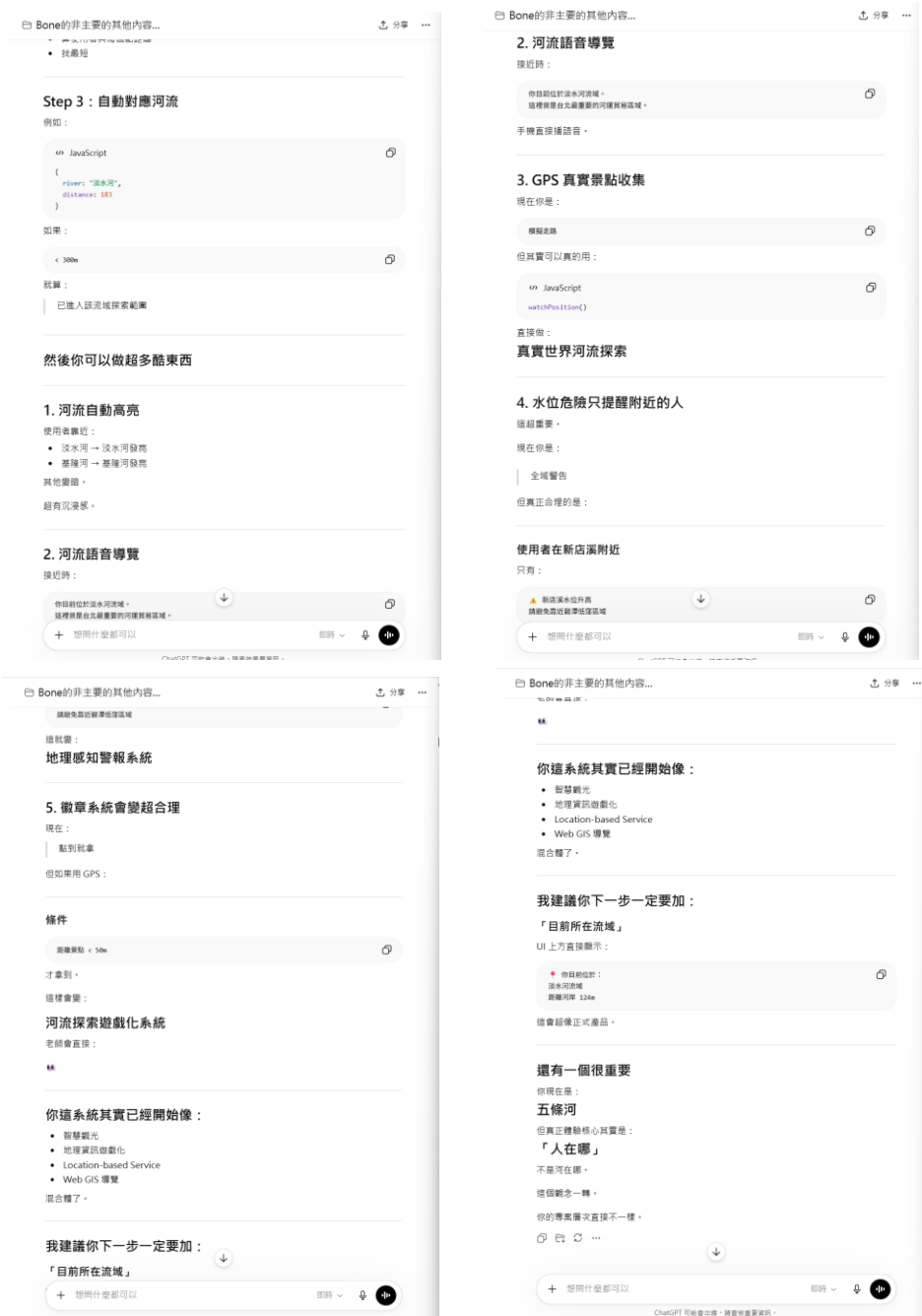
M0-LB03	Quiz score ranking
M0-LB04	Badge ranking
M0-LB05	River exploration ranking
M0-LB06	Display name visibility toggle



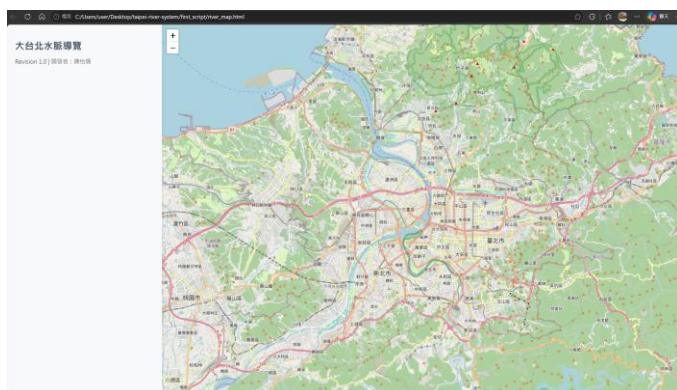
附錄：

1、作品著作者：陳怡珊、葉庭蓁、生成式 AI





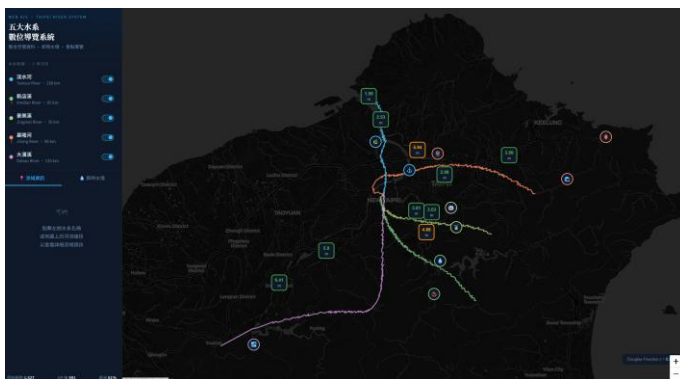
(2) 下載後長這樣，但好像不太行，所以又問 AI



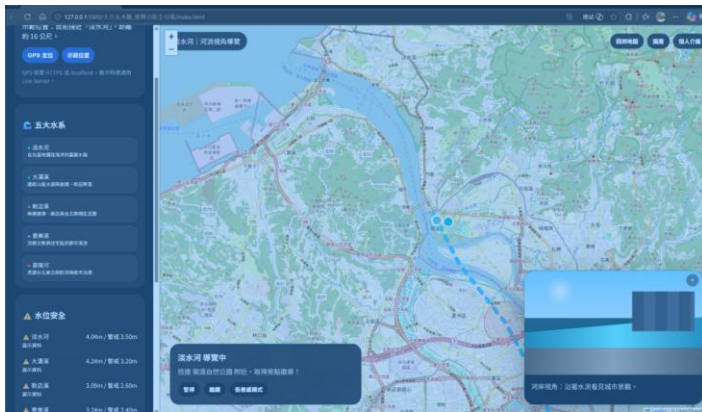
(2) 他和我說不能幫我弄後端，所以我讓他幫我把前端生出來看看



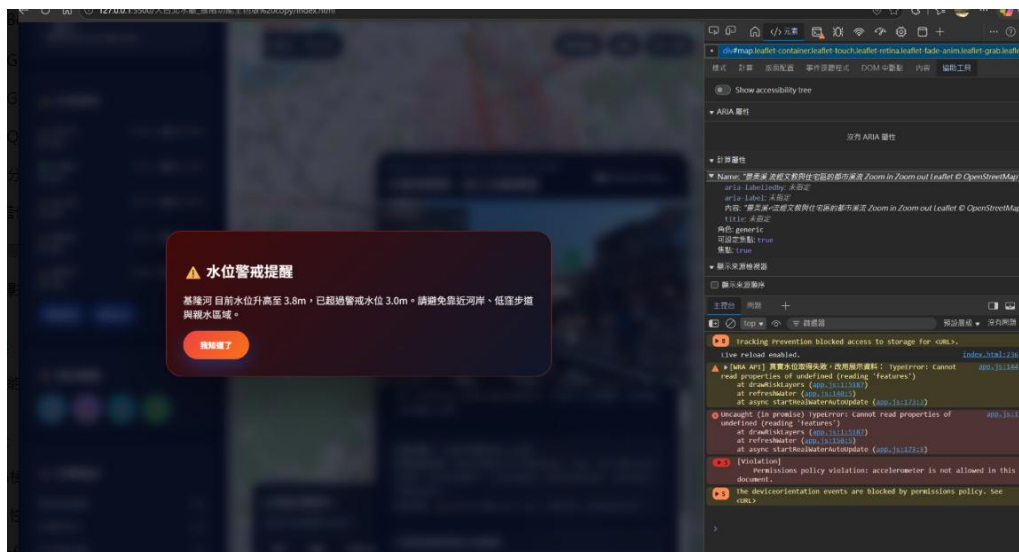
(3)然後這是 AI 生給我的畫面的二版



(4) 我又繼續問，讓他再生第三版，但我發現他不會串景觀的資料，所以我只好繼續問他，自己加入街景資料...。



(5) 監測站的資料他拉不到，所以我只好自己查監測站位置然後拉進去 JSON 裡面。





Bone的非主要的其他內容...

你現在找到問題核心了：
你的 API 資料檔也根本不是：
而是政府那種怪異命名法：
而且你現在對 `jsoncode.html`：
可能結構像：`id`：
代表：
✅ 建議字元查詢規則
❌ 但像像 `pickstationname()` / `pickrivername()` 沒找到正確檔位
所以匹配流程中迷失掉。

直接修 (這次是關鍵)
找 `pickstationname`
把整段改成：

```
function pickstationname(m) {
  return String(
    getJsonValue(
      {
        "observatoryname",
        "observatoryname",
        "stationname",
        "stationname",
        "chinese",
        "chinese",
        "english",
        "english",
        "locationdescription",
        "location",
        "name",
        "name"
      },
      m
    )
  );
}
```

Bone的非主要的其他內容...

直接修 (這次是關鍵)
找 `pickrivername`
把整段改成：

```
function pickrivername(m) {
  return String(
    getJsonValue(
      {
        "rivername",
        "rivername",
        "observatoryname",
        "observatoryname",
        "stationname",
        "stationname",
        "chinese",
        "chinese",
        "english",
        "english",
        "locationdescription",
        "location",
        "name",
        "name"
      },
      m
    )
  );
}
```

Bone的非主要的其他內容...

然後最重要
找：
把：
把整段改成：
再補強關鍵字
把：
把整段改成：

```
function pickstationname(m) {
  return String(
    getJsonValue(
      {
        "observatoryname",
        "observatoryname",
        "stationname",
        "stationname",
        "chinese",
        "chinese",
        "english",
        "english",
        "locationdescription",
        "location",
        "name",
        "name"
      },
      m
    )
  );
}
```

Bone的非主要的其他內容...

基礎河也補
把整段改成：
為什麼你現在會失敗
因為政府資料根本不是：
而是：
是結構：
你程式：
把整段改成：
把整段改成：

```
function pickstationname(m) {
  return String(
    getJsonValue(
      {
        "observatoryname",
        "observatoryname",
        "stationname",
        "stationname",
        "chinese",
        "chinese",
        "english",
        "english",
        "locationdescription",
        "location",
        "name",
        "name"
      },
      m
    )
  );
}
```

Bone的非主要的其他內容...

直接修 (這次是關鍵)
找 `pickstationname`
把整段改成：

```
function pickstationname(m) {
  return String(
    getJsonValue(
      {
        "observatoryname",
        "observatoryname",
        "stationname",
        "stationname",
        "chinese",
        "chinese",
        "english",
        "english",
        "locationdescription",
        "location",
        "name",
        "name"
      },
      m
    )
  );
}
```

Bone的非主要的其他內容...

已思考 25%
你現在其實在思考，問題在這樣：
Console 已經找到：
但 `Array()` 代表 測試資料有錯誤，但 `stationid` 沒對上即時水位資料。
你這樣寫正確寫法 ID 的檔位是：
你現在在 `pickstationname()` 沒找到 `match` 不到，你的 `app.js` 目前還是用 `observatoryname` / `stationid` 之類檔位在。把下面整段粘到 `app.js` 最底部：

```
function pickstationname(m) {
  return String(
    getJsonValue(
      {
        "observatoryname",
        "observatoryname",
        "stationname",
        "stationname",
        "chinese",
        "chinese",
        "english",
        "english",
        "locationdescription",
        "location",
        "name",
        "name"
      },
      m
    )
  );
}
```

Bone的非主要的其他內容...

把整段改成：

```
function pickstationname(m) {
  return String(
    getJsonValue(
      {
        "observatoryname",
        "observatoryname",
        "stationname",
        "stationname",
        "chinese",
        "chinese",
        "english",
        "english",
        "locationdescription",
        "location",
        "name",
        "name"
      },
      m
    )
  );
}
```

Bone的非主要的其他內容...

把整段改成：

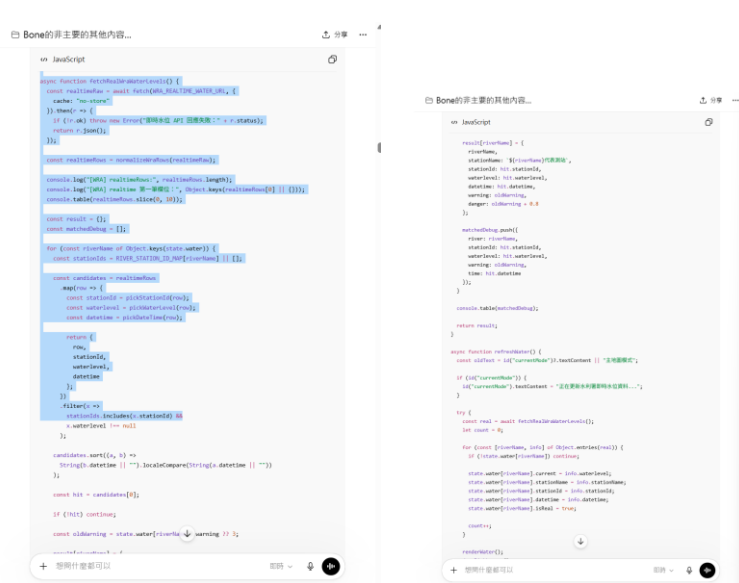
```
function pickstationname(m) {
  return String(
    getJsonValue(
      {
        "observatoryname",
        "observatoryname",
        "stationname",
        "stationname",
        "chinese",
        "chinese",
        "english",
        "english",
        "locationdescription",
        "location",
        "name",
        "name"
      },
      m
    )
  );
}
```

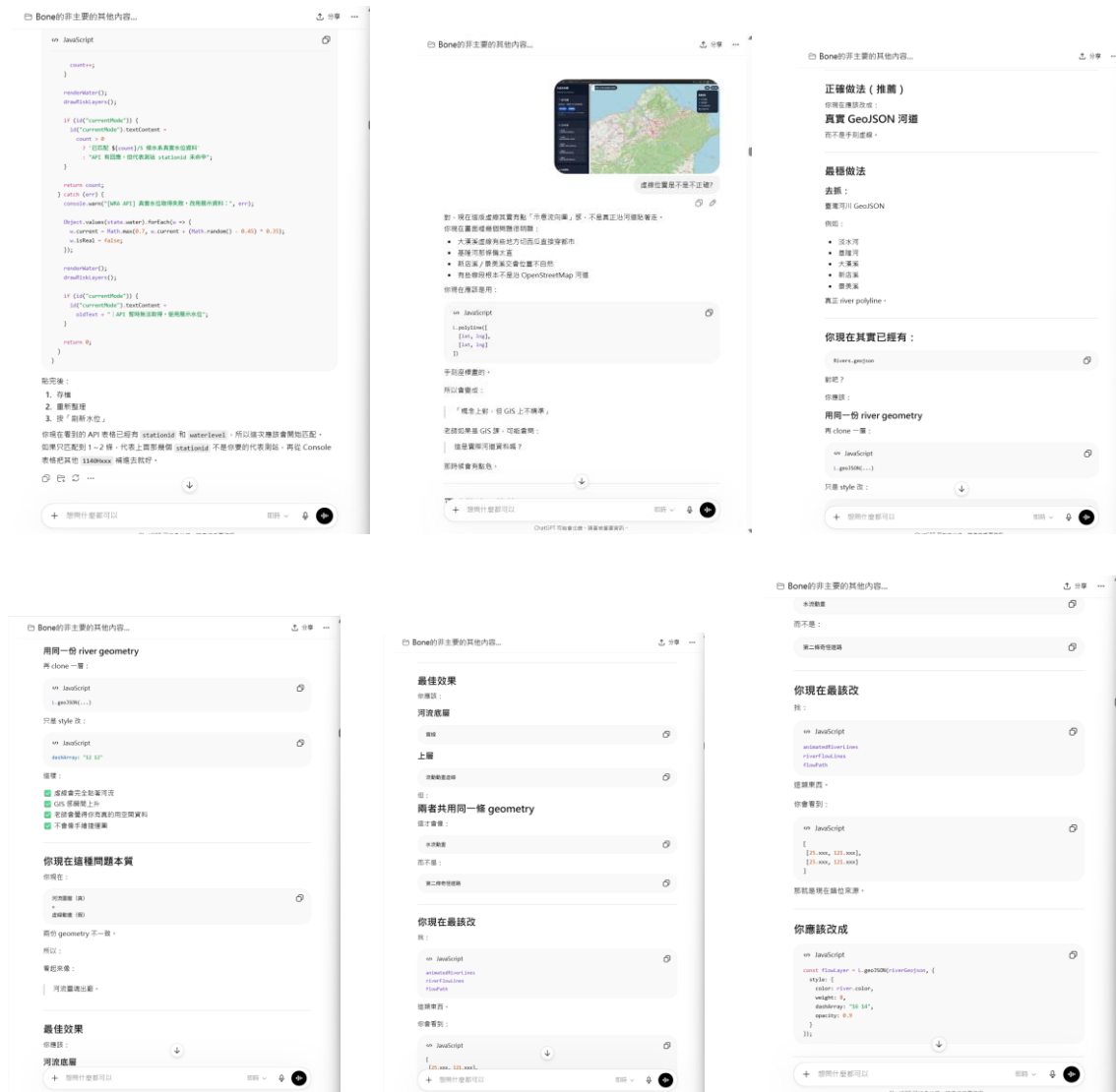
Bone的非主要的其他內容...

已思考 30%
你現在問題不是檔位沒加，而是 測試資料的 ID 跟即時水位資料的 ID 沒有直接對上。你現在 `app.js` 裡有非常多重複格式，但最後用的檔案前面，所以直接在最前面再粘兩段。
把下面整段粘到 `app.js` 最底部：

```
function pickstationname(m) {
  return String(
    getJsonValue(
      {
        "observatoryname",
        "observatoryname",
        "stationname",
        "stationname",
        "chinese",
        "chinese",
        "english",
        "english",
        "locationdescription",
        "location",
        "name",
        "name"
      },
      m
    )
  );
}
```







(5)之後是我們人類自己改河流座標做轉換，調整程式碼然後加入其他景點，加資料庫變成現在這個版本。